



USN

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First/Second Semester B.Tech (CBCS) Degree Examinations
Summer Semester July 2024

MA1007-1- DISCRETE MATHEMATICS AND TRANSFORM TECHNIQUES
(AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from **each Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The logical consequence of the statement $[(\sim p \vee \sim q) \wedge q]$ is:
A) $\sim q$ B) $\sim p$
C) p D) q
2. Contrapositive statement of the statement $\neg p \rightarrow \neg q$ is:
A) $\neg p \rightarrow \neg q$ B) $p \rightarrow q$
C) $\neg q \rightarrow p$ D) $q \rightarrow p$
3. If $P(x)$: "the word x contains the letter a ", then which of the following statement is false:
A) $P(\text{track})$ B) $P(\text{true})$
C) $P(\text{false})$ D) $P(\text{orange})$
4. Which of the following is true:
A) $\neg \forall x P(x) \equiv \forall x \neg P(x)$ B) $\neg \forall x P(x) \equiv \exists x \neg P(x)$
C) $\neg \exists x P(x) \equiv \exists x \neg P(x)$ D) $\neg \forall x P(x) \equiv \exists x P(x)$
5. Which of the following relation is the non-empty relation on A , where A is a set of positive integers not exceeding 3.
A) $R = \{(a, b) | b = a^6\}$ B) $R = \{(a, b) | b = a + 3\}$
C) $R = \{(a, b) | b = \frac{1}{4}a\}$ D) $R = \{(a, b) | b = 10a\}$
6. If graph G has 7 vertices with degrees 2, 4, 6, 5, 7, 8, 10, then number of edges in graph G is:
A) 62 B) 42
C) 31 D) 21
7. Let $M_R = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ be the matrix of the relation R on A , then the relation R is:
A) Symmetric, reflexive but not transitive B) Transitive, reflexive but not symmetric
C) Equivalence relation D) Symmetric, transitive but not reflexive
8. Which of the following graph is non-planar?
A) $K_{3,3}$ B) W_5
C) K_4 D) C_7
9. The root of the equation $x^3 + x - 1 = 0$ lies in the interval _____
A) (1, 2) B) (0, 1)
C) (2, 3) D) (4, 5)
10. The method which approximates curve as tangent to find root of an algebraic equation is:
A) Newton-Raphson method B) Regula-Falsi method
C) Runge-Kutta method D) Modified Euler's method

11. In Runge –Kutta method of order four , the formula to find k_1 is:
 A) $k_1 = f(x_0, y_0)$ B) $k_1 = h f(x_0, y_0)$
 C) $k_1 = \frac{h}{2} f(x_0, y_0)$ D) $k_1 = f\left(x_0 + \frac{h}{2}, y_0\right)$
12. The standard five point formula for solving Laplace's equation $u_{xx} + u_{yy} = 0$ is:
 A) $4 u_{i,j} = u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}$ B) $3 u_{i,j} = u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1} - u_{0,0}$
 C) $2 u_{i,j} = u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1}$ D) $u_{i,j} = u_{i-1,j} + u_{i+1,j} + u_{i,j-1} - u_{i,j+1}$
13. In the Fourier series expansion of $f(x)$ defined in the interval $(-c, c)$, the coefficient b_n is:
 A) $\frac{1}{c} \int_{-c}^c f(x) dx$ B) $\frac{1}{c} \int_{-c}^c f(x) \cos\left(\frac{n\pi x}{c}\right) dx$
 C) $\frac{1}{c} \int_{-c}^c f(x) \sin\left(\frac{n\pi x}{c}\right) dx$ D) $\int_{-c}^c f(x) \sin\left(\frac{n\pi x}{c}\right) dx$
14. The Half range Fourier cosine series expansion of the function $f(x)$ in $(0, \pi)$ is:
 A) $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$ B) $\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\pi x)$
 C) $\sum_{n=1}^{\infty} a_n \cos(nx)$ D) $\sum_{n=1}^{\infty} b_n \sin(nx)$
15. If $F(\omega)$ is the complex Fourier transforms of the function $f(t)$, then the Fourier transform of $f(5t)$ is:
 A) $\frac{1}{\omega} F\left(\frac{\omega}{5}\right)$ B) $\frac{1}{5} F\left(\frac{\omega}{5}\right)$
 C) $F\left(\frac{\omega}{5}\right)$ D) $5F\left(\frac{\omega}{5}\right)$
16. If $F(\omega)$ is the complex Fourier transforms of the function $f(t)$, then the Fourier transform of $f(t - 6)$ is:
 A) $e^{-i\omega} F(\omega - 6)$ B) $e^{-6i\omega} F(\omega - 6)$
 C) $e^{-i(\omega-6)} F(\omega)$ D) $e^{-6i\omega} F(\omega)$
17. Let u_n be a two sided sequence. If $Z(u_n) = U(z)$ with the ROC $r_1 < |z| < r_2$, then $Z(u_{-n})$ is :
 A) $U\left(\frac{1}{z}\right)$ with the ROC $\frac{1}{r_2} < |z| < \frac{1}{r_1}$ B) $U(2z + 4)$ with the ROC $\frac{1}{r_1} < |z| < \frac{1}{r_2}$
 C) $U(z^2)$ with the ROC $r_1 < |z| < \frac{1}{r_1}$ D) $U(z^3)$ with the ROC $\frac{1}{r_2} < |z| < r_2$
18. If u_n is a function of n then the bilateral Z –transform of u_n is defined as, $Z(u_n) =$
 A) $\sum_{n=0}^{\infty} u_n z^n$ B) $\sum_{n=0}^{\infty} u_n z^n$
 C) $\sum_{n=-\infty}^0 u_n z^{-n}$ D) $\sum_{n=-\infty}^{\infty} u_n z^{-n}$
19. If u_n, v_n, w_n are any discrete functions, then $Z(6u_n + 5v_n - 3w_n)$ is
 A) $3 Z(u_n) + 6Z(v_n) - 3 Z(w_n)$ B) $-6 Z(u_n) - 5Z(v_n) + 3Z(w_n)$
 C) $6 Z(u_n) + 5Z(v_n) - 3Z(w_n)$ D) $6 Z(u_n) 5Z(v_n) 3Z(w_n)$
20. $Z(a^{nt})$ where a, t are constants is:
 A) $\frac{z}{z+a^t}$ B) $\frac{z}{z-a^t}$
 C) $\frac{z}{z+a^{-t}}$ D) $\frac{z}{z-a^{-t}}$

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

1. a) (i) Show that the statement $p \wedge (q \vee r) \leftrightarrow (p \wedge q) \vee (p \wedge r)$ is a tautology by using truth table.
 (ii) Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.
- b) Use mathematical induction to show that
 $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n .
- c) Determine the truth value of the following statements, if the domain of each variable consists of all real numbers.
 (i) $\neg \forall x (x^2 > 2)$ (ii) $\neg \exists x, x \leq x + 1$.

Marks	BT*	CO*	PO*
6	L*3	1	2
6	L1	1	1
4	L2	1	1

2. a) Write the following argument in symbolic form and then establish the validity of the argument using the rules of inference. "If Dominic goes to the racetrack, then Helen will be mad". "If Ralph plays cards all night, then Carmela gets mad". "If either Helen or Carmela gets mad, then their lawyer Veronica will be notified". "Veronica has not heard from either of these clients". Consequently, "Dominic did not make it to the racetrack and Ralph didn't play cards all night".

6 L3 1 2

6 L2 1 2

4 L1 1 1

- b) Prove that $\sqrt{2}$ is irrational.

- c) Show that $\neg \forall x (P(x) \rightarrow Q(x))$ and $\exists x (P(x) \wedge \neg Q(x))$ are logically equivalent.

Unit – II

3. a) Let $A = \{1, 2, 3, 4, 6\}$. Consider the relation R on A defined by aRb if and only if " a divides b ".

(i) Find the relation R .

(ii) Find the matrix of R and draw the digraph of R .

(iii) Find in degree and out degree of each element of A .

(iv) List all the paths of length two from the vertex 2 in the digraph of R .

6 L1 2 1

- b) (i) Define graph colouring.

(ii) Schedule the final exams for Math 115, Math 116, Math 185, Math 195, CS 101, CS 102, CS 273, and CS 473, using the fewest number of different time slots, if there are no students taking both Math 115 and CS 473, both Math 116 and CS 473, both Math 195 and CS 101, both Math 195 and CS 102, both Math 115 and Math 116, both Math 115 and Math 185, and both Math 185 and Math 195, but there are students in every other pair of courses.

6 L3 2 2

- c) Let $A = \{1, 2, 3, 4, 5, 6\}$ and $p = (1, 4) \circ (2, 3, 5)$ be a permutation of A . Compute p^{-1} and $p \circ p$. Write the result as the product of disjoint cycles.

4 L2 2 1

4. a) Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 1), (2, 2), (3, 1), (3, 4), (4, 3)\}$. Find the transitive closure of the relation R by using Warshall's algorithm.

6 L2 2 2

- b) Define adjacency matrix. Draw the graphs corresponding to each of the following adjacency matrices.

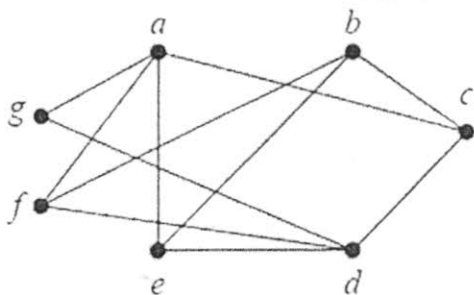
$$G = \begin{bmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{bmatrix}$$

$$H = \begin{bmatrix} 0 & 2 & 3 & 0 \\ 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 0 \\ 1 & 0 & 0 & 2 \end{bmatrix}$$

6 L1 2 1

- c) i) Define bipartite and complete bipartite graphs.

(ii) Check whether the following graph is bipartite. Justify your answer.



4 L3 2 1

Unit – III

5. a) Find a real root of the equation $x^3 - 2x - 5 = 0$ in the interval $(2, 3)$, using regula-falsi method. Carry out three iterations. 6 L2 3 1
- b) Solve the equation $u_{xx} + u_{yy} = 0$ under the conditions $(h = 1, k = 1)$, $u(0, y) = 0$, $u(3, y) = 3 + y$, for $0 \leq y \leq 3$; $u(x, 0) = x$, $u(x, 3) = 2x$ for $0 \leq x \leq 3$. 6 L3 3 1
- c) Use Taylor's series method to find the value of y at $x = 0.1$, given that $\frac{dy}{dx} = x + y$, $y(0) = 1$, take $h = 0.1$. 4 L2 3 2
6. a) Use Runge-Kutta fourth order formula to find $y(0.1)$, given that $y' = x + y^2$, $y(0) = 1$, take $h = 0.1$. 6 L2 3 1
- b) Solve the boundary value problem $u_t = u_{xx}$ under the conditions $u(0, t) = u(1, t) = 0$ and $u(x, 0) = \sin \pi x$, $0 \leq x \leq 1$ using Bendre-Schmidt relation (take $h = 0.2$, $k = 0.02$ and $\alpha = \frac{1}{2}$) for $0 \leq t \leq 0.06$. 6 L3 3 1
- c) Find a real root of the equation $x^4 - x - 10 = 0$, using Newton-Raphson method near $x_0 = 2$. Carry out two iterations. 4 L2 3 2

Unit – IV

7. a) Express $f(x) = x - x^2$ as a Fourier series in the interval $-\pi < x < \pi$. 6 L2 4 2
- b) Find the Fourier series expansion of the function $f(x) = \begin{cases} -k & -\pi < x < 0 \\ k & 0 < x < \pi \end{cases}$ and $f(x + 2\pi) = f(x)$. 6 L2 4 2
- c) Find the half range Fourier sine series of $f(x) = 2x$ in $(0, 2)$. 4 L2 4 1
8. a) Prove that $F\left(\int_{-\infty}^{\infty} f(u) g(t - u) du\right) = F\{f(t)\} \cdot F\{g(t)\}$. 6 L3 4 2
- b) Find the Fourier transform of $f(t) = \begin{cases} 1 & \text{for } |t| \leq 1 \\ 0 & \text{for } |t| > 1 \end{cases}$. Hence evaluate $\int_0^{\infty} \frac{\sin t}{t} dt$. 6 L3 4 2
- c) Show that $F_S\{t f(t)\} = -\frac{d}{d\omega}\{F_C(\omega)\}$ and $F_C\{t f(t)\} = \frac{d}{d\omega}\{F_S(\omega)\}$. 4 L2 4 1

Unit – V

9. a) Find the z-transform and ROC(region of convergence) of $u_n = \begin{cases} 5^n & \text{if } n < 0 \\ 3^n & \text{if } n \geq 0 \end{cases}$. 6 L2 5 1
- b) State and prove convolution theorem for bilateral z-transforms. 6 L2 5 2
- c) Find the inverse bilateral z-transform of $U(z) = \frac{1}{(z-2)(z-3)}$. 4 L3 5 1
10. a) State and prove final value theorem of unilateral z-transforms. 6 L2 5 1
- b) Solve the following difference equation by using z-transforms $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$, $u_0 = 1$ and $u_1 = 1$. 6 L3 5 2
- c) Find unilateral z-transform of the following functions:
- (i) $u_n = \left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n$
- (ii) $u_n = 2n + 5 \sin\left(\frac{n\pi}{4}\right) - 3a^2$. 4 L2 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
